Nicole Keeney

Geospatial data engineer and full stack developer building climate data tools

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SUMMARY

Build data infrastructure, web tools, and open source software for Cal-Adapt, California's climate data platform: processing climate model output on HPC and AWS, cataloging it with STAC, and building the websites and maps people use to explore it. Trained as an atmospheric scientist, with graduate research on global flood impacts.

EDUCATION

M.S., Civil & Environmental Engineering, Colorado State University 2025
B.A., Atmospheric Science, University of California, Berkeley 2020

EXPERIENCE

Eagle Rock Analytics Remote · Jun 2022 – Present
Research Software Engineer, 2026 – Present
Scientific Developer (part time), 2024 – 2025
Junior Atmospheric Scientist, 2022 – 2023

- Build data pipelines that process downscaled climate projections on HPC and AWS into cloud-optimized zarr, producing climate hazard datasets like extreme heat and heat wave metrics.
- Manage Cal-Adapt's STAC ecosystem, from data ingestion to the STAC API and STAC Browser, using an open standard to make climate datasets searchable.
- Develop interactive climate data tools for the Cal-Adapt: Data Explorer in Next.js and TypeScript, and maintain the tile server that delivers gridded climate layers to its maps.
- Lead technical development of the Cal-Adapt: Analytics Engine website, which provides guidance and resources for using California climate data.
- Core contributor to climakitae, Cal-Adapt's open source Python package for accessing and analyzing climate data.
- Led cloud workflows on AWS ParallelCluster for quality control and standardization of weather data from 1,400+ stations.

Colorado State University Fort Collins, CO · Aug 2023 – Dec 2025
Graduate Research Assistant, Department of Civil & Environmental Engineering

- Built a global dataset of flood events from 2000 to 2024 by combining disaster records with satellite imagery, then used it to study how extreme precipitation drives flood impacts.
- Ran the processing on HPC and presented the work at the AGU Fall Meeting 2025.

University of Maryland / NASA Goddard Space Flight Center Remote · Jun 2020 – Jan 2022
Faculty Research Assistant, half time · NASA Summer Intern

- Contributed to a cloud-optimized Python toolkit for validating polar climate models with satellite data on Google Cloud.
- Built an interactive Jupyter Book evaluating drivers of winter Arctic sea ice growth using NASA ICESat-2 data.

UC Berkeley School of Public Health Berkeley, CA · Oct 2020 – Feb 2022
Junior Specialist, half time · Undergraduate Student Researcher

- Developed Python and R workflows for environmental data analysis supporting public health research, and calibrated a wind erosion model using satellite vegetation data.

SKILLS

Data engineering: Python, xarray, zarr, Dask, geopandas, rasterio, STAC (pystac, pgSTAC, stac-fastapi), intake-esm
Full stack: TypeScript, React, Next.js, FastAPI, PostgreSQL, MapLibre, AI-assisted development
Cloud & HPC: AWS (S3, Lambda, RDS, SAM, ParallelCluster), Docker, Slurm, GitHub Actions
Climate science: Climate projections, climate hazard analysis, flood mapping, remote sensing